

B. Tech. Examination, 2024-25

(Odd Semester)

ARTIFICIAL INTELLIGENCE IN MECHANICAL ENGINEERING SYSTEMS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION-A

1. Attempt all parts of the following :

8×1=8

- (a) What is an end effector?
- (b) Define machine learning.
- (c) What is actuator?
- (d) What is fixed routing?
- (e) Define automation.

[P. T. O.]

- (f) What is CAPP?
- (g) Define AI in banking sector.
- (h) Define tachometer.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Explain the five levels of automation and control in manufacturing.
- (b) What is the role of artificial intelligence in thermal powerplants? Explain in detail.
- (c) Describe knowledge acquisition and how it is useful in artificial intelligence system.
- (d) Explain the types of joints with diagrams.

SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

- 3. (a) Explain error detection and recovery and types of error recovery.
- (b) Define analog-digital converters.

- (c) What is an expert system?
4. (a) Explain the working of articulated robot and SCARA with diagram.
- (b) Define MRP and explain how it works?
- (c) How does adaptive control work?
5. (a) Define actuators according to the type of drive power.
- (b) Differentiate between machine learning and data mining.
- (c) Discuss the role of AI in manufacturing sector.
6. (a) Define the following :
- (i) Tactile sensors
 - (ii) Proximity sensors
 - (iii) Optical sensors
- (b) Define the architecture of expert system.
- (c) Define additive manufacturing and how does 3D